

# StatefulSymbolicContext#

[https://pytorch.org/docs/stable/generated/torch.fx.experimental.symbolic\\_shapes.StatefulSymbolicContext.html](https://pytorch.org/docs/stable/generated/torch.fx.experimental.symbolic_shapes.StatefulSymbolicContext.html)

## Description:

Create symbols in `create_symbolic_sizes_strides_storage_offset` via a `symbolic_context` determination as given by a cache of `Source:Symbol`. A cache hit will reuse a stored symbol, and a cache miss will write to this cache. This behaves like `StatelessSymbolicContext`, except the cache supersedes the other values - `dynamic_sizes` and `constraint_sizes` will not be read if we cache hit. It is the cache owner's responsibility to maintain the lifecycle of the cache with respect to different `shape_envs`, clearing, etc.

## Function Signature

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```
class
torch.fx.experimental.symbolic_shapes.StatefulSymbolicContext(
    dynamic_sizes, dynamic_strides=None,
    constraint_sizes=None, constraint_strides=None,
    specialize_on=None, view_base_context=None,
    tensor_source=None,
    shape_env_to_source_to_symbol_cache=None)[source]#
```

# Detailed Documentation

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## Documentation

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This behaves like `StatelessSymbolicContext`, except the cache supersedes the other values - `dynamic_sizes` and `constraint_sizes` will not be read if we cache hit.

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